

SOUTHWEST HAWAII BUOY 51002, USA

WMO: 992430

Lat: 17.043N

Lon: 157.742W

Elev: 0

StdP: 101.33

Time Zone: -11.00 (W11)

Period: 94-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	22.2	22.7	N/A	N/A	N/A	N/A	N/A	N/A	14.2	23.9	13.6	23.8	8.3	80	0.743

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9	1.1	28.2	N/A	27.8	N/A	27.4	N/A	N/A	N/A	N/A	N/A	N/A	N/A	6.7	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.2	11.7	11.0	DB	20.6	27.4	0.9	1.1	20.0	28.2	19.4	28.9	18.9	29.5	18.2	30.4	(4)
(5)			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec			
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)			
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	25.0	24.2	23.9	24.0	24.2	24.6	25.1	25.6	26.0	26.1	26.0	25.5	24.7	(6)	
(7)		DBStd	1.06	0.68	0.71	0.67	0.71	0.78	0.75	0.69	0.74	0.75	0.68	0.67	0.66	(7)	
(8)		HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
(9)		HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)	
(10)		CDD10.0	5475	439	389	436	427	452	454	483	495	484	496	465	455	(10)	
(11)		CDD18.3	2433	181	156	177	177	194	204	224	237	234	238	215	197	(11)	
(12)		CDH23.3	16138	747	542	679	768	1053	1417	1787	2071	2123	2110	1688	1153	(12)	
(13)	CDH26.7	327	0	0	0	0	3	3	27	75	120	78	21	1	(13)		
(14)	Wind	WSAvg	8.0	7.4	7.7	8.1	8.4	7.9	8.1	8.0	7.4	7.9	8.4	8.5	(14)		
(15)	Precipitation	PrecAvg	459	45	45	49	19	14	10	23	42	46	54	54	57	(15)	
(16)		PrecMax	898	143	170	202	78	89	56	64	150	215	187	130	194	(16)	
(17)		PrecMin	205	0	1	3	5	1	1	1	3	5	10	16	4	(17)	
(18)		PrecStd	158	41	43	51	20	19	10	17	35	42	39	30	48	(18)	
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	26.3	25.8	25.8	26.2	27.0	27.1	27.8	28.4	29.0	28.1	27.8	26.8	(19)	
(20)			MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(20)
(21)		2%	DB	25.8	25.7	25.7	25.8	26.7	26.9	27.3	27.9	28.5	27.8	27.3	26.4	(21)	
(22)			MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(22)
(23)		5%	DB	25.5	25.4	25.3	25.6	26.3	26.7	27.0	27.6	28.0	27.4	26.9	26.1	(23)	
(24)			MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(24)
(25)	10%	DB	25.2	25.2	25.2	25.4	25.8	26.4	26.8	27.0	27.4	27.2	26.7	25.8	(25)		
(26)		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(27)	
(28)			MADB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(28)
(29)		2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(29)
(30)			MADB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(30)
(31)		5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(31)
(32)			MADB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(32)
(33)	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(33)	
(34)		MADB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	1.3	1.3	1.2	1.1	1.2	1.2	1.1	1.1	1.2	1.3	1.4	(35)		
(36)			MCDBR	1.4	1.5	1.3	1.3	1.6	1.3	1.3	1.3	1.5	1.5	1.7	1.8	(36)	
(37)			MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(37)
(38)		5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(38)	
(39)			MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.359	0.369	0.377	0.406	0.400	0.390	0.392	0.393	0.384	0.378	0.375	0.367	(40)		
(41)		taud	2.605	2.580	2.558	2.446	2.484	2.504	2.518	2.525	2.547	2.601	2.598	2.603	(41)		
(42)		Ebn at Noon	931	937	935	901	895	894	894	901	914	916	908	912	(42)		
(43)		Edh at Noon	93	99	104	117	111	108	107	107	104	96	93	91	(43)		
(44)	All-Sky Solar Radiation	RadAvg	4.98	5.45	6.06	6.67	7.00	7.05	7.05	6.88	6.53	5.69	4.90	4.55	(44)		
(45)		RadStd	0.24	0.30	0.54	0.40	0.33	0.16	0.22	0.26	0.22	0.20	0.26	0.28	(45)		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page